

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0714

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

Annexure A

Industry Problem-Solving Task

1. A star topology is ideal for a 20-person startup because all devices connect to a central switch, making the network easy to manage and troubleshoot. At the core sits a Layer 2/Layer 3 managed switch (such as a Cisco Catalyst or Netgear ProSafe) with at least 24 ports to accommodate all workstations, printers, and VoIP phones with room to grow. Each device connects to the switch using Cat6 UTP cables, which support Gigabit speeds up to 100 meters — more than sufficient for any typical office floor plan. The switch uplinks to a router/firewall (such as a Cisco RV series or pfSense appliance) that handles internet access, DHCP, and network address translation. A Wi-Fi access point connected to the switch extends wireless coverage for laptops and mobile devices.

The key advantage of this design is fault isolation: if one workstation's cable fails, only that single node goes offline, leaving all other 19 employees unaffected. Centralised cable management also simplifies future moves, adds, and changes.

2. A bank demands near-zero downtime, making a full or partial mesh topology the right choice. In a full mesh, every core network device — routers, core switches, and servers — connects directly to every other device, so no single link or node failure can bring down the network. For a bank, a partial mesh is typically deployed at the core layer (connecting all core switches and firewalls to each other) while the distribution and access layers use redundant uplinks to at least two core switches. Dual enterprise-grade routers (such as Cisco ASR or Juniper MX series) provide BGP-based redundant internet connectivity. All links between core devices use fiber optic cables — either single-mode for long distances between buildings or multi-mode OM4 within the data center — because fiber is immune to electromagnetic interference, supports very high bandwidth, and is essential for maintaining low-latency financial transactions. Redundant power supplies, UPS systems, and RAID storage arrays further complement the mesh design. The justification is straightforward: when a link fails in a mesh, traffic automatically reroutes through alternative paths in milliseconds using protocols like OSPF or BGP, satisfying regulatory requirements for uptime and data integrity.

3. For a small classroom lab with a tight budget, bus topology offers the lowest hardware cost because all computers share a single common communication line — the bus — eliminating the need for a central switch or hub. A single coaxial cable (RG-58, 50-ohm) or, in a modern variation, a single Cat5e/Cat6 cable daisy-chained through an inexpensive unmanaged hub runs along the length of the room. Each computer connects to the backbone using a T-connector and a BNC connector in a classic coaxial setup, with terminators (50-ohm) placed at both ends of the cable to absorb signals and prevent reflection. For a more contemporary low-cost implementation, a single inexpensive 8- or 16-port unmanaged switch replaces the terminators and uses Cat5e UTP patch cables to each workstation — achieving bus-like simplicity at minimal cost. The primary limitation is that the network is susceptible to collisions and a single cable break disrupts all connected machines, but for a supervised classroom environment with light usage (internet browsing, file sharing), this trade-off is entirely acceptable. No dedicated server is required; a teacher's PC can share resources using built-in Windows/Linux file sharing.
4. A large university campus serves thousands of users across multiple buildings — lecture halls, labs, libraries, dormitories, and administration — making a single topology inadequate. A hybrid topology combines the best elements of multiple designs: a fiber-based ring or mesh at the campus backbone level for redundancy, a star topology within each building for manageability, and bus or

tree structures within individual labs for cost-efficiency. At the campus core, high-speed fiber optic cables (single-mode, 10 Gbps or 40 Gbps) interconnect core routers and multilayer switches using a ring or partial mesh arrangement, ensuring that no single link failure isolates an entire building. Each building has its own distribution switch connected to the campus core via dual fiber uplinks. Within each building, a traditional star topology branches from the distribution switch to floor-level access switches, which then connect individual devices via Cat6 cables. This hierarchical three-layer model — core, distribution, and access — is the industry standard for enterprise and campus networks because adding a new building simply means installing a new distribution switch and running fiber to the core, with no disruption to existing infrastructure. Wireless access points throughout the campus connect via PoE (Power over Ethernet) to access switches, eliminating the need for separate power cabling. The hybrid design thus delivers redundancy at the backbone, simplicity at the building level, and scalability campus-wide.

5. Designing a network for a company with multiple departments involves creating a structured and scalable architecture using switches and routers to ensure efficient communication and security. Each department (such as HR, Finance, Sales, and IT) is connected through dedicated Layer 2 switches that form the access layer, allowing devices like computers and printers to communicate within the same department. These switches are then

connected to a central Layer 3 switch or router, which acts as the core layer and enables communication between different departments through inter-VLAN routing. Virtual LANs (VLANs) are configured to logically separate each department's network, improving security and reducing broadcast traffic. A router is also used to connect the internal network to external networks such as the internet, often with firewall functionality for security. Redundant links and backup paths can be included to ensure reliability, while proper IP addressing and subnetting are applied for efficient network management. This hierarchical design improves performance, enhances security, and allows easy expansion as the company grows.

1. Design a suitable network topology for a 20-person startup and justify your choice.

Introduction

A 20-person startup typically operates from a single floor or small office and needs a network that is inexpensive to set up, easy for a small (or non-existent) IT team to manage, and tolerant of individual device failures. Among the classic topologies — bus, ring, star, mesh, tree and hybrid — the star topology is the most appropriate choice, and this answer justifies that choice with device selection, cabling, logical design, cost, and risk analysis.

Proposed Design

- Core device: one Layer 2/Layer 3 managed switch (e.g., Cisco Catalyst 1000 series or Netgear ProSafe), 24–48 ports, Gigabit Ethernet, to allow room for growth.
- Edge/Internet device: a router/firewall appliance (Cisco RV340, pfSense, or a UTM device) providing NAT, DHCP, VPN and stateful firewall inspection.
- Cabling: Cat6 UTP horizontal cabling (100 m max run) from the switch to every workstation, printer and VoIP handset, terminated using the TIA/EIA-568B standard.
- Wireless: one or two 802.11ax (Wi-Fi 6) access points connected to the switch via PoE, covering laptops, tablets and BYOD devices.
- Logical segmentation: two VLANs — Staff/Data and Guest/Wi-Fi — to isolate visitor traffic from internal resources.
- Backup connectivity: a secondary low-cost broadband/4G failover link on the router for basic redundancy at low added cost.

Physical Layout (described)

All twenty end devices are wired individually back to a central switch mounted in a small comms cabinet or patch panel, forming the classic 'hub-and-spoke' pattern of a star topology. The switch itself uplinks to the router, which connects to the ISP modem/ONT. Because every cable run terminates at one point, structured cabling with a patch panel keeps the office tidy and each drop individually labelled and testable.

Justification

Star topology is justified on four grounds. First, fault isolation: a single failed cable, NIC, or patch port only removes that one workstation from the network — the other nineteen employees are unaffected, unlike a bus topology where a single break can down the whole segment. Second, ease of management: adding, moving or removing an employee's PC only involves plugging into a free switch port and, if VLANs are used, assigning the port to the correct VLAN — no re-cabling of the whole network is required. Third, performance: Gigabit Ethernet with a switch gives each device a dedicated collision-free link, unlike shared-medium topologies, so twenty users experience no meaningful contention for typical office traffic (email, browsing, file sharing, VoIP). Fourth, cost-effectiveness: for only twenty devices, a single 24–48-port managed switch and Cat6 cabling is inexpensive relative to the cost of a mesh or ring design, while still leaving spare ports for growth.

Limitations and Mitigation

- Single point of failure at the central switch — mitigated by choosing a reputable business-grade switch, keeping a cold-spare switch on hand, and covering it under next-business-day vendor support.
- Cabling cost grows with distance from the switch — mitigated by placing the switch centrally in the office rather than at one end.

- No inherent path redundancy — acceptable for a 20-person startup where downtime of minutes is tolerable, unlike a bank or hospital.

Conclusion

For a small, budget-conscious, easy-to-manage office, a switch-centred star topology with Cat6 structured cabling, VLAN segmentation and a small wireless overlay delivers the best balance of cost, manageability, performance and fault isolation, making it the recommended design for a 20-person startup.

2. Propose a topology for a bank requiring near-zero downtime and justify your choice.

Introduction

Banking networks carry real-time financial transactions, ATM/POS traffic, core banking application data and regulatory reporting feeds. Any outage translates directly into lost revenue, regulatory penalties and reputational damage, so the network must be engineered for near-zero downtime. This requires a topology built around redundancy at every layer: physical links, devices, power, and routing paths. A (partial) mesh topology at the core, combined with a redundant hierarchical design elsewhere, is the correct choice.

Proposed Design

Layer	Design	Redundancy Mechanism
Core	Partial/full mesh of core switches & firewalls	Every core device has ≥ 2 direct links to other core devices
Distribution	Redundant switches, dual uplinks	Each distribution switch links to two different core switches
Access	Access switches with dual uplinks	Spanning-Tree/ VSS or stacking prevents loops while keeping redundancy
WAN/Internet	Dual ISPs via BGP, diverse physical entry points	Automatic failover if one carrier or path fails
Power	Dual PSUs, UPS, backup generator	No single power event stops the network
Storage	RAID arrays, replicated data centres	Data survives disk or site failure

- Core devices: enterprise-class routers such as Cisco ASR or Juniper MX series, configured in redundant pairs and running BGP for internet redundancy and HSRP/VRRP for gateway redundancy.
- Transmission media: fiber optic cabling throughout — single-mode fiber for inter-building/inter-site links, multi-mode OM4 fiber within the data centre — chosen for immunity to electromagnetic interference, very high bandwidth and low latency, which is essential for real-time transaction processing.
- Routing protocols: OSPF or BGP with fast-convergence tuning (e.g., BFD) so that if a link fails, traffic reroutes within milliseconds rather than the tens of seconds of default timers.
- Redundant data centres: an active-active or active-passive secondary data centre in a different geographic location, connected via diverse fiber paths, so a site-level disaster does not take the bank offline.

Justification

A mesh design is justified because it removes the single point of failure that a star, bus or ring topology would introduce. In a full mesh, every core node connects directly to every other node, so the failure of any one link or device still leaves alternative paths available; a partial mesh applies this principle economically by fully meshing only the small number of critical core devices while distribution and access layers use dual-homed (redundant) uplinks rather than a full mesh, which would be prohibitively expensive at scale. This matches the way real banks and other mission-critical networks are built, and it satisfies regulatory expectations (e.g., PCI-DSS, central-bank IT-resilience guidelines) for continuity of service. Fiber optic media further reduces the number of possible faults — no EMI-induced errors, no copper-length restrictions — while dynamic routing (OSPF/BGP) automatically detects and routes around failures without manual intervention, achieving 'near-zero' rather than theoretically impossible 'zero' downtime.

Cost and Trade-offs

Mesh topology is the most expensive design to build and maintain because of the number of links, ports and cabling required ($n(n-1)/2$ links for a full mesh of n nodes), and it needs skilled staff to design and troubleshoot the routing. This cost is justified for a bank because the cost of downtime — fraud exposure, SLA penalties, customer trust — far outweighs the capital and operational expenditure of the redundant design.

Conclusion

A partial-mesh core over redundant fiber links, combined with dual-homed distribution/access layers, dual ISPs, redundant power and a geographically separated backup data centre, gives the bank the automatic, sub-second failover needed to approach near-zero downtime while keeping the design economically realistic.

3. Recommend a low-cost topology for a small classroom lab and justify your choice.

Introduction

A small classroom computer lab is a low-risk, budget-constrained, supervised environment: traffic is light (web browsing, document editing, occasional file sharing), downtime of a few minutes is not critical, and the priority is minimising the cost of hardware and cabling per student PC. This makes bus topology — or its modern low-cost equivalent, a single inexpensive unmanaged switch — the recommended design.

Proposed Design (Classic Bus)

- Backbone: a single RG-58 (50-ohm) coaxial cable, or a daisy-chained Cat5e cable, run along the length of the classroom.
- Connectors: each PC taps into the backbone using a T-connector/BNC connector (coaxial) or an inexpensive switch port (modern variant).
- Terminators: 50-ohm terminating resistors at both physical ends of the coaxial cable to absorb signal and prevent reflection/echo.
- No dedicated server: a teacher's PC shares files/printers using built-in OS file-sharing (Windows/Linux), avoiding the cost of a server.

Proposed Design (Modern Low-Cost Equivalent)

In practice, true coaxial bus networks are obsolete, so the recommended low-cost modern implementation is a single inexpensive 8- or 16-port unmanaged switch (no licensing, no configuration, roughly the cost

of one to two workstations) with Cat5e patch cables running to each PC. Logically this still behaves like a shared, single-collision-domain segment from a cost perspective — one shared device, minimal cabling complexity — while gaining full-duplex, collision-free performance because a switch (unlike a hub) does not actually share bandwidth the way a bus does.

Justification

Bus-style (or single-unmanaged-switch) design is justified purely on a cost-per-port and simplicity basis: it needs no central intelligent device, no VLAN configuration, no routing, and minimal cable — a single run instead of individually homed cables to a wiring closet. For twenty to thirty student PCs in one room used for light, supervised tasks, this is by far the cheapest way to give every machine internet and local file access. Installation and maintenance require no networking expertise, which matters in a school setting without dedicated IT staff.

Limitations and Why They Are Acceptable Here

Limitation	Impact	Why acceptable in a classroom lab
Single cable/segment failure	Whole lab loses connectivity	Lab is supervised; a short outage is a minor inconvenience, not a financial loss
Collisions on true coax bus	Slower performance under heavy load	Typical classroom load (browsing, light file transfer) rarely saturates the segment
No redundancy	No automatic failover	Redundant hardware is not cost-justified for a teaching lab
Limited scalability	Hard to extend cleanly	Lab size is fixed and known in advance

Conclusion

Given the classroom lab's priorities — lowest possible cost, minimal management overhead, and tolerance for occasional short outages — a bus topology (or its practical modern form, a single unmanaged switch with Cat5e cabling) is the most suitable and economical recommendation.

4. Design a network for a large university campus with multiple buildings and justify your choice.

Introduction

A large university campus hosts thousands of users across many buildings — lecture halls, laboratories, libraries, dormitories, sports facilities and administration — each with different traffic patterns, security needs and device densities. No single flat topology can meet these requirements, so the recommended design is a hybrid topology built on the industry-standard three-layer hierarchical model: core, distribution, and access.

Proposed Design

Layer	Function	Devices / Media
Core	High-speed campus backbone connecting all buildings; ring/partial-mesh for redundancy	Core routers & multilayer switches; single-mode fiber, 10–40 Gbps links
Distribution	Aggregates each building's traffic; applies routing, VLAN and policy boundaries	Layer 3 distribution switches; dual fiber uplinks to two core devices
Access	Connects end devices on each floor/lab	Layer 2 access switches; Cat6 UTP to desktops, PoE to Wi-Fi APs and IP phones

- Backbone: fiber optic ring or partial mesh linking core routers/switches in each major building, so a single fiber cut does not isolate any building — traffic simply flows the other way around the ring or via an alternate mesh path.
- Per-building distribution: a Layer 3 distribution switch in each building with dual fiber uplinks to two different core devices for redundancy, and inter-VLAN routing for that building's departments.
- Per-floor access: star-wired access switches on each floor connect to end devices using Cat6 (Gigabit, up to 100 m); this is the 'star within the hybrid' referenced by the question.
- Wireless: campus-wide 802.11ax access points powered and connected via PoE from access switches, using a WLAN controller for centralised management and seamless roaming between buildings.
- Security & segmentation: VLANs per department/faculty, a campus firewall/NGFW at the internet edge, NAC (network access control) for student and staff device onboarding, and a separate VLAN/SSID for guest/student residential traffic.

Justification

A hybrid design is justified because it applies the right topology at the right scale: mesh/ring redundancy where an outage would be catastrophic (the backbone linking entire buildings), star simplicity where manageability matters most (wiring within a building or lab), and a layered hierarchy that scales cleanly — adding a new building means installing one new distribution switch and running fiber to the core, without redesigning the rest of the campus. High-capacity single-mode fiber on the backbone supports the aggregate bandwidth of thousands of simultaneous users (video lectures, research data transfer, cloud services) with headroom for growth, while PoE for access points removes the need for separate electrical wiring to every AP location. This three-layer hierarchical model is the de facto industry standard (as used by Cisco's Enterprise Campus Architecture) precisely because it has been proven at university and enterprise scale.

Scalability and Resilience

Because each layer has a clearly defined role, the design scales horizontally (new buildings, new floors, new access switches) without re-engineering existing layers, and it scales in resilience terms because a fault at the access layer affects only one floor, a fault at the distribution layer affects only one building (which itself has dual uplinks), and the ring/mesh core ensures the backbone itself has no single point of failure.

Conclusion

A hybrid, three-layer hierarchical topology — fiber ring/mesh core, redundant star distribution per building, and star access per floor — delivers the redundancy, performance, security segmentation and manageable scalability that a large multi-building university campus requires.

5. Design a scalable network architecture for a company with multiple departments and justify your choice.

Introduction

A company with multiple departments (HR, Finance, Sales, IT, Operations, etc.) needs a network that keeps each department's traffic logically separated for security and performance, while still allowing controlled inter-department communication and centralized internet access, and that can grow as the company adds staff or departments. The recommended architecture is a hierarchical switched network using VLANs and inter-VLAN routing, layered as access, core/distribution, and edge.

Proposed Design

- Access layer: a dedicated Layer 2 access switch (or switches) per department (HR, Finance, Sales, IT), connecting each department's PCs and printers via Cat6 cabling.
- VLAN segmentation: each department is assigned its own VLAN (e.g., VLAN 10 = HR, VLAN 20 = Finance, VLAN 30 = Sales, VLAN 40 = IT) so that broadcast traffic and Layer 2 visibility are confined within a department, improving both security and performance.
- Core/distribution layer: a central Layer 3 switch (or router-on-a-stick for a smaller company) that performs inter-VLAN routing, allowing controlled communication between departments (e.g., Finance to IT for support tickets) according to access-control-list (ACL) policy.
- Edge/Internet layer: a router with integrated firewall connects the internal network to the internet, applying NAT and inbound/outbound filtering; a DMZ VLAN hosts any public-facing services (web/email servers).
- Redundancy: dual uplinks from each access switch to the core switch (or a redundant core pair) with Spanning Tree Protocol / Rapid PVST+ to prevent loops while allowing a backup path if one uplink fails.
- IP addressing: structured subnetting per VLAN (e.g., 10.10.x.x for HR, 10.20.x.x for Finance) using VLSM to use address space efficiently and to keep routing tables simple.

Justification

This architecture is justified primarily on security and scalability grounds. VLANs enforce logical separation between departments on shared physical switching hardware, so a compromised or misbehaving device in Sales cannot flood broadcast traffic into Finance's segment, and access-control lists on the Layer 3 core can restrict which departments are allowed to reach sensitive resources (e.g., only HR and Finance can reach the payroll server VLAN). The hierarchical layering also directly supports scalability: adding a new department means creating a new VLAN and connecting a new access switch to the existing core — no redesign of the rest of the network is required — while adding more users within a department means adding another access switch trunked to the same VLAN. Centralising routing and internet access at the core/edge means security policy, NAT and firewalling are managed in one place rather than scattered across the network.

Benefits Summary

Requirement	How the design meets it
Departmental security	Per-department VLANs isolate broadcast domains; ACLs control inter-VLAN access
Performance	Smaller broadcast domains per VLAN reduce unnecessary traffic on each segment
Scalability	New department = new VLAN + access switch; no re-cabling of existing departments
Manageability	Centralized inter-VLAN routing, NAT and firewall policy at the core/edge
Reliability	Redundant uplinks and STP prevent a single link failure from isolating a department

Conclusion

A VLAN-segmented, hierarchical switched architecture — departmental access-layer switches, a Layer 3 core for inter-VLAN routing and policy, and a secured edge for internet connectivity — gives the company the security isolation, manageable growth path and reliable performance that a scalable multi-department network requires.